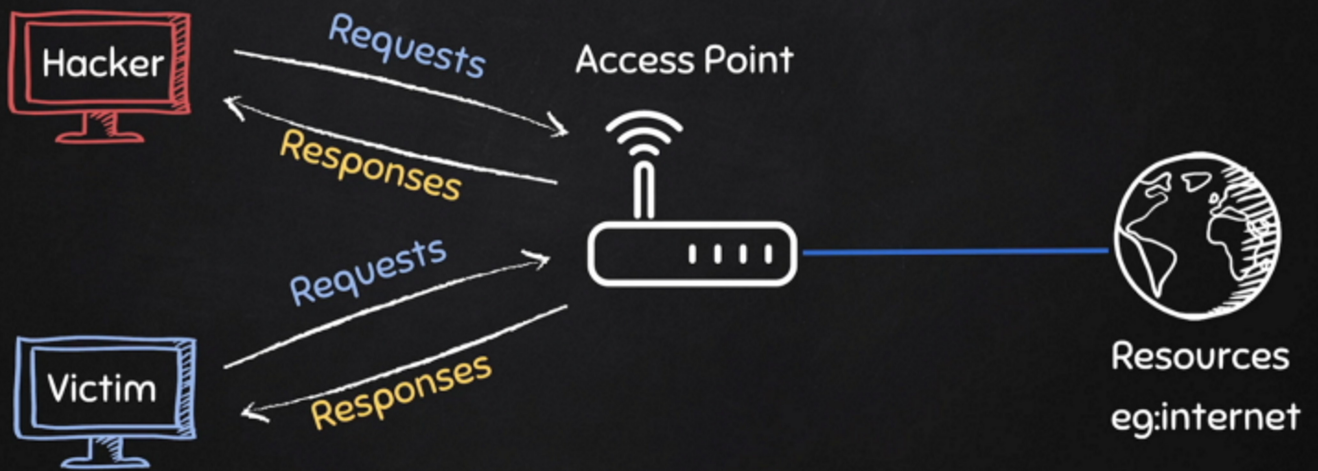
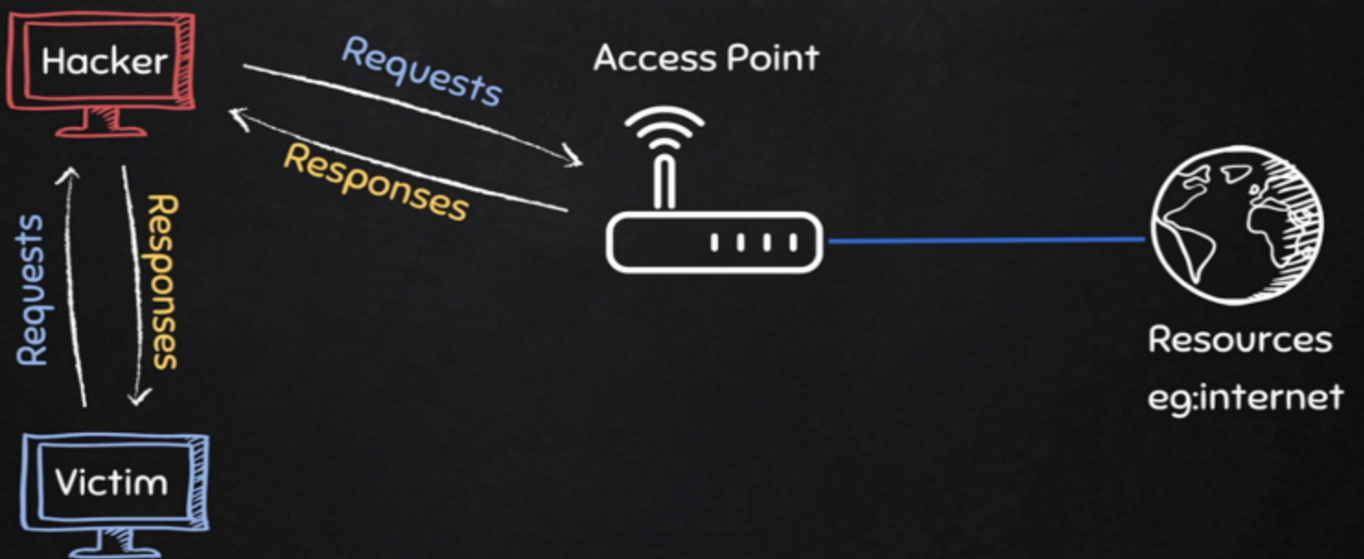


TYPICAL NETWORK



ARP SPOOFING





What is ARP Spoofing? (Explained Simply)

ARP Spoofing is a type of cyber attack where an attacker tricks devices on a **local network (LAN)** into sending their data to the **attacker's machine**, instead of the **real gateway/router**.



What is ARP?

ARP (Address Resolution Protocol) is used to map IP addresses (e.g., 192.168.1.1) to **MAC addresses** (hardware addresses like AA:BB:CC:DD:EE:FF) within a local network.

When a device wants to send data, it asks:

“Who has IP 192.168.1.1? Tell me your MAC address.”

The real gateway replies:

“I do. My MAC is AA:BB:CC:DD:EE:FF.”



In ARP Spoofing

The **attacker** sends **fake ARP replies** like:

“Hey! I’m 192.168.1.1 — my MAC is FF:FF:FF:FF:FF:FF.”

Now, the victim **believes** the attacker is the gateway. So the victim **sends all traffic** (HTTP, passwords, etc.) to the attacker.



Purpose of ARP Spoofing

1. **Man-in-the-Middle (MITM)** attack
 - Attacker intercepts traffic between victim and gateway
2. **Data sniffing** (stealing passwords, sessions)
3. **Session hijacking** (e.g., stealing login cookies)
4. **SSL Stripping** (removing encryption from HTTPS sites)
5. **Denial of Service (DoS)**



ARP Spoofing Tools in Kali Linux

Tool	Use Case
bettercap	Full MITM suite with SSL strip
arp spoof	Basic ARP spoofing from dsniff
ettercap	GUI and CLI MITM attack suite
mitmf	Advanced MITM attack framework